Exam: labeling software

Github repo

https://github.com/Timofejs-Rimensons/Labeling-Software

Problem statement

- I want to make labeling **storefronts & signage** as efficient as possible. UI should be specialised for these types of objects and be easy to navigate for untrained users.

Research

- The UI in my case is made for labeling **storefronts & signage** which have some distingtive properties as:
  - Majority of the objects are racktangles, observed from random angles. (vertical angle changes less than horisontal)
  - Projection of these objects on the screen is mostly trapezoids.
  - Using bounding box is the eaciest trivial method for labeling such data.
    - The alternative is selecting corner nodes of objects to accurately capture trapezoids.
- Survey was collected verbaly amonst CTAI students. It included the next question:
  - Check out these examples of data. What method would you prefer for labeling **storefronts & signage** - bounding box or selecting nodes?

- The majority (2 of 3) people chose nodes selection as the labeling method. That means that in my UX and UI I'll be using this method for demonstration.

Value proposition

A labeling tool purpose-built for **storefronts and signage** that lets any labeler, trained or not, accurately capture trapezoid annotations using 4-node corner

selection. Faster than generic tools, more accurate than bounding boxes, and simple enough to be used by anyone.

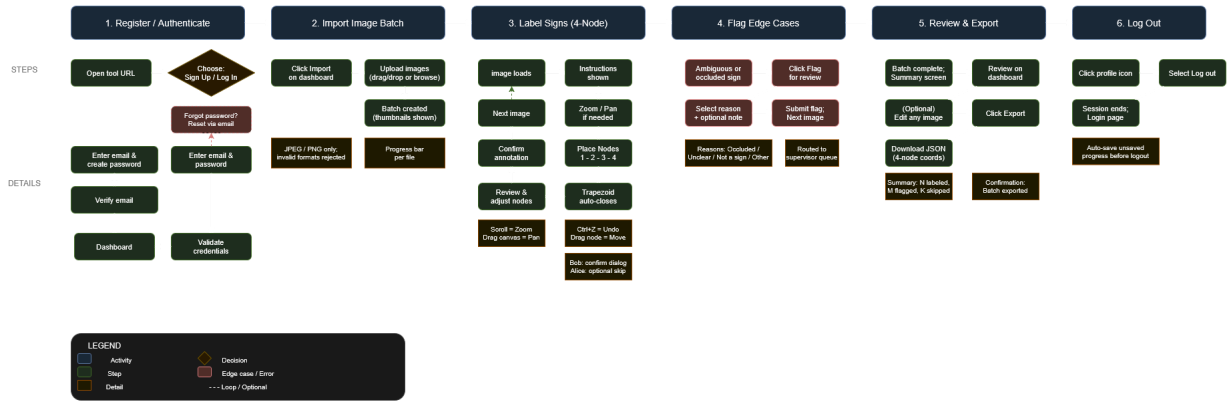
User personas

- Alice
  - Is a student with course focused on AI.
  - Is familiar with other labeling apps.
  - Has an inside on what labeled data should look like and what to do in edge cases.
- Bob
  - Is a gig worker in data labeling company.
  - Is familiar with other labeling apps.
  - Doesn’t have an inside on how to solve edge cases, needs client supervision.

User Stories

- As **Alice**, I want to place exactly 4 corner nodes on a sign so that the annotation accurately captures its trapezoid shape in perspective.
- As **Alice**, I want to see my nodes highlighted so that I can quickly review and correct them before submitting.
- As **Bob**, I want flagging option to mark unresolved images.
- As **any user**, I want to zoom in on an image so that I can accurately place nodes on small or distant signs.
- As **any user**, I want to undo and reposition a node so that I can correct mistakes without restarting.
- As **any user**, I want visual feedback when 4 nodes are placed so that I know the annotation is complete.

User flows



Project scope

In scope:

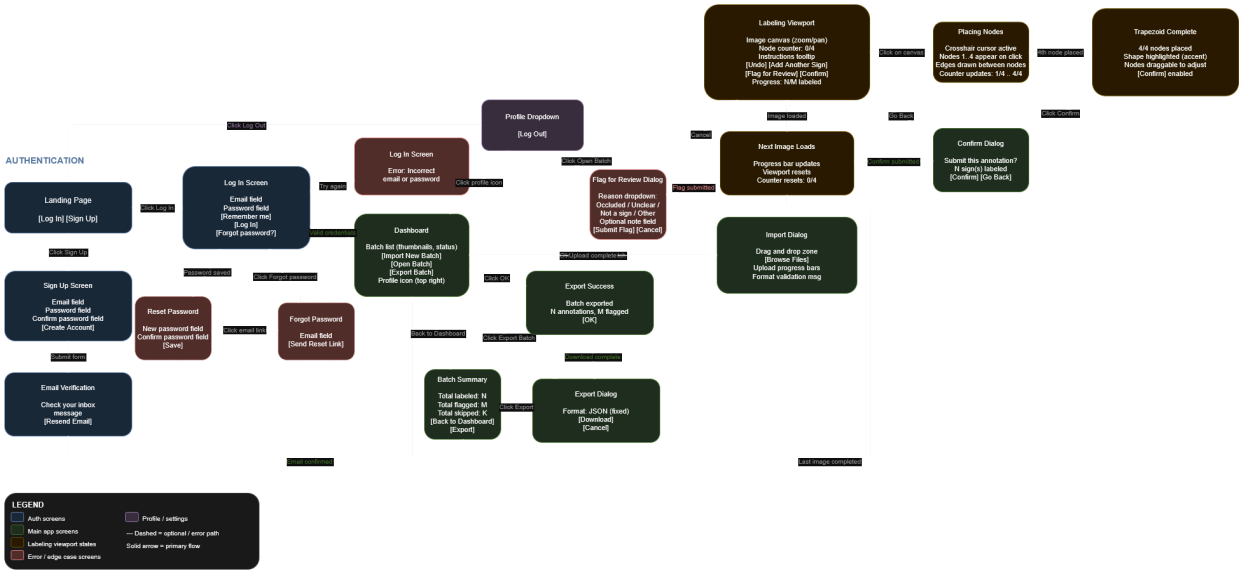
- Node-based trapezoid annotation tool (4-corner selection).
- UI optimised for storefront and signage object types.
- Support for untrained users.
- Images import/export.
- Basic labeling workflow: load images → place nodes → confirm label → next image.

Out of scope:

- Generic polygon annotation (non-4-corner shapes).
- Bounding box mode (excluded based on survey results and geometric reasoning).
- AI-assisted pre-labeling or auto-detection.
- Export format configuration (assumed fixed for this project).

- Mobile support.

Competitors

Type	Price	Upsides	Drawbacks
Labelme	Open-source	Free	Polygon annotation; lightweight; JSON export; large community
CVAT	Open-source / Cloud	Free / Paid	Multi-shape tools; task management; team support; active development
Label Studio	Open-source + Enterprise	Free / Paid	Configurable templates; ML backend; review workflows; REST API

Screen relations